Section 4.1 Restricted permutations & arrangements

受限排列

Method 1 / 方法 1 [CO01-M01]

Separate the choice of a type pattern from the assignment of distinct labels. Product rules need a fixed number of legal continuations; quotient rules need a proof of equal multiplicity. Restrictions often become simpler after choosing an extreme element, a run, or a cycle decomposition.

先分离类型模式与不同标签的安排。乘法原理须保证合法后继数确定，归并除法须证明每个对象被数同样多次。固定极值元素、连续段或循环分解后，限制常会简化。

Multiply sequential choices after identifying the state

明确状态后逐步相乘

Procedure

1. State what has already been chosen and what remains available.

2. Count legal choices at each step, checking whether the count depends on the earlier history.

3. Multiply only when these continuation counts are valid for every allowed history; sum disjoint process lengths or states separately.

通用步骤

1. 写清已选内容与尚可使用的内容。

2. 逐步计算合法选择数，并检验是否依赖此前具体历史。

3. 只有所有合法历史均具有所用后继数时才相乘；过程长度或状态不同则分类相加。

> Example. Choose an ordered three-station tour from five distinct stations without repeats. At each visited station choose one of two activities. How many tours with activities are there?

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Method 1 / 方法 1 [CO01-M01] (continued / 续)

> 例题：从五个不同站点选不重复的三站顺序游览，每站选两种活动之一，共有多少个带活动安排的行程?

Worked solution

1. Station choices are $5, 4, 3$, independent of which earlier stations were used.

2. Every station choice has 2^3 activity assignments.

3. The total is $5 \cdot 4 \cdot 3 \cdot 2^3 = 480$; the activity choice does not remove stations.

例题解析

1. 站点选择依次为五、四、三，与具体已选站点无关。

2. 每个站点顺序有 2^3 种活动安排。

3. 总数为 $5 \cdot 4 \cdot 3 \cdot 2^3 = 480$，活动选择不减少后续站点。

A condition with a uniform predecessor count may be easier to build right-to-left than left-to-right.

若前驱数恒定，从右向左构造可能比从左向右简单。

If a two-symbol local rule forbids only UD, every legal word is D^aU^b, so its length-n patterns number $n+1$. This structural split replaces an unnecessary long recurrence. In a substitution process, the number currently on the field may stay fixed while the unused incoming pool shrinks; distinguish these two counts.

若双符号局部规则只禁 UD，每个合法串均为 D^aU^b，长度 n 的模式数为 $n+1$，无须建立长递推。换人过程可能保持场上人数固定而未使用替补池缩小，须区分这两个选择数。

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Method 1 / 方法 1 [CO01-M01] (continued / 续)

Practice / 配套练习: AIME-2003-II-Q03 (WB p. 613), AIME-2005-I-Q05 (WB p. 615), AIME-2016-I-Q03 (WB p. 694), AIME-2019-I-Q04 (WB p. 620)

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Method 2 / 方法 2 [CO01-M02]

Choose positions when the internal order is forced

内部顺序被迫时只选位置

Procedure

1. Identify the element forced to be an extreme, such as the minimum at a valley.

2. Choose which labels go on each monotone side or type class.

3. Place each class in its forced order; for alternating inequalities, count admissible rank patterns before assigning values.

通用步骤

1. 找出被迫为极值的元素，例如谷底的最小值。

2. 选择各单调侧或类型类所用的标签。

3. 按被迫顺序放置各类；若为交错不等式，先数合法排名模式再赋实际值。

> Example. Merge three alphabetically ordered mathematics titles and three alphabetically ordered art titles into one shelf, preserving each list order. Count the shelves.

> 例题：将各三本已按字母顺序排列的数学书、艺术书合并成一排，分别保持原列表顺序，求排法数。

Worked solution

1. Choose the three of six shelf positions occupied by mathematics titles.

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Method 2 / 方法 2 [CO01-M02] (continued / 续)

2. The mathematics list must fill those slots in its given order; the art list then fills the other slots in its given order.

3. This is a bijection with three-subsets of six positions, giving $\binom{6}{3} = 20$.

例题解析

1. 选择六个位置中哪三个放数学书。

2. 数学书按原顺序填这些位置，艺术书按原顺序填其余位置，均无额外选择。

3. 所以与六位置的三元子集双射，共 $\binom{6}{3} = 20$ 种。

A forced monotone order contributes one arrangement, not a factorial.

被迫单调的顺序只贡献一种，不能再乘阶乘。

Practice / 配套练习: AIME-2004-I-Q06 (WB p. 625), AIME-2006-II-Q04 (WB p. 616), AIME-2017-I-Q11 (WB p. 796)

Section 4.1 Restricted permutations & arrangements

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Method 3 / 方法 3 [CO01-M03]

Count multiplicity patterns with multinomial weights

用多项式系数加权重数模式

Procedure

1. Solve the total-count and weighted-total equations for all possible multiplicity vectors.

2. For a vector $(n_1, \ldots, n_r)$, count its words by $N!/\prod n_i!$.

3. If only some symbols repeat, split by their used multiplicities before choosing and ordering the other symbols.

4. For unlabeled groups or unordered entries, remove exceptional repeated cases before dividing by their permutation multiplicity.

通用步骤

1. 解总数量与加权总量方程，列全可行重数向量。

2. 重数为 $(n_1, \ldots, n_r)$ 的串有 $N!/\prod n_i!$ 个。

3. 若只有部分符号可重复，先按实际使用重数分类，再选排其余符号。

4. 无标识组或无序项应先处理重复例外，再按排列重数归并。

> Example. Six positions contain three red, two blue, and one green marker, identical within a color. How many words have green before every blue?

> 例题：六个位置放三红、二蓝、一绿同色不可区分标记，要求绿标记在所有蓝标记之前，求串数。

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Method 3 / 方法 3 [CO01-M03] (continued / 续)

Worked solution

1. First choose the three positions occupied by nonred markers in $\binom{6}{3}$ ways.

2. In those positions the order must be green, blue, blue; the three red positions are also forced.

3. The answer is 20. Equivalently, among the $6!/(3!2!) = 60$ unrestricted words, green has three equiprobable ranks among the nonred markers.

例题解析

1. 先选三个非红位置，有 $\binom{6}{3}$ 种。

2. 这些位置只能依次为绿、蓝、蓝，红位置也全部确定。

3. 故答案为二十；亦可从无限制的 $6!/(3!2!) = 60$ 串中按绿色在非红中的三个等可能排名取一。

The same symbol-count vector may carry a different probability if individual outcome probabilities differ.

若各类单次概率不同，相同重数向量的概率还须乘相应幂权重。

Practice / 配套练习: AIME-2001-II-Q11 (WB p. 789), AIME-2004-II-Q04 (WB p. 614), AIME-2007-II-Q01 (WB p. 617), AIME-2008-I-Q09 (WB p. 786), AIME-2009-I-Q09 (WB p. 650), AIME-2016-II-Q08 (WB p. 476), AIME-2018-II-Q08 (WB p. 674), AIME-2018-II-Q15 (WB p. 655), AIME-2019-I-Q05 (WB p. 730), AIME-2021-I-Q04 (WB p. 648), AIME-2025-I-Q03 (WB p. 624)

Section 4.1 Restricted permutations & arrangements

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Method 4 / 方法 4 [CO01-M04]

Glue runs, place gaps, then fill labels

先粘连续段、放间隔，再填标签

Procedure

1. Turn each required consecutive group into a block, retaining its possible internal orders.

2. Place blocks in a line, a circle, or gaps of the other objects; check the endpoints and empty-gap rules.

3. For forbidden adjacency, count selected forced-adjacency blocks inside inclusion-exclusion.

4. Only after the color or group pattern is fixed, assign distinct members within each group.

通用步骤

1. 将须连续的每组粘成区块，并保留内部可排顺序。

2. 在线、圆或其他物体间隔中放区块，核对两端与空间隔规则。

3. 禁相邻时，在容斥内部计算所选强制相邻边产生的区块。

4. 先固定颜色或群体模式，再把不同成员填入各组位置。

> Example. Six distinct people A, B, C, D, E, F stand in a row. The first three named people must be consecutive, while D, E must be separated. Count the rows.

> 例题：六个不同的人 A, B, C, D, E, F 排成一行，A, B, C 必须连续，D, E 不得相邻，求排法数。

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Method 4 / 方法 4 [CO01-M04] (continued / 续)

Worked solution

1. Glue A, B, C into one internally ordered block. With D, E, F, there are $4! \cdot 3! = 144$ rows.

2. Among these, glue D, E too: three outer blocks have $3!$ orders, the first block has $3!$ internal orders, and D, E have two orientations, giving 72.

3. Subtract to obtain $144 - 72 = 72$. The two blocks use disjoint people, so their counts combine directly.

例题解析

1. 把 A, B, C 粘成可内部排列的区块，与 D, E, F 共四个单位，有 $4! \cdot 3! = 144$ 种。

2. 其中再把 D, E 粘成块，外部三块有 $3!$ 顺序，第一块内部 $3!$，二人方向两种，共七十二种。

3. 相减得七十二种，两块人员不交，因此其选择可相乘。

A circular block needs a specified anchor unless the rotation action has already been accounted for.

圆形区块须有指定锚点，除非已明确处理旋转归并。

Practice / 配套练习: AIME-2000-II-Q05 (WB p. 647), AIME-2009-I-Q10 (WB p. 651), AIME-2011-I-Q12 (WB p. 765), AIME-2011-II-Q07 (WB p. 652), AIME-2011-II-Q12 (WB p. 745), AIME-2014-II-Q09 (WB p. 663), AIME-2018-I-Q03 (WB p. 728), AIME-2019-II-Q05 (WB p. 621), AIME-2020-II-Q09 (WB p. 665), AIME-2025-II-Q10 (WB p. 653)

Section 4.1 Restricted permutations & arrangements

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Method 5 / 方法 5 [CO01-M05]

Decompose permutations into cycles and track symmetry

将置换分解为循环并追踪对称

Procedure

1. Translate an iterate, invariant-subset, or degree-two graph condition into allowed cycle lengths.

2. List the cycle-length partitions of the total size.

3. For a permutation with m_j cycles of length j, use $N!/\prod_j (j^{m_j} m_j!)$; undirected cycles have an extra reversal identification.

4. Divide a labeled circular arrangement by its rotation count only after proving there are no nontrivial rotational stabilizers.

通用步骤

1. 将迭代、不变子集或二度图条件转成允许循环长度。

2. 列出总大小的循环长度分拆。

3. 长度 j 循环有 m_j 个的置换数为 $N!/\prod_j (j^{m_j} m_j!)$；无向圈还须处理反向归并。

4. 有标识圆排列按旋转归并前，先证明没有非平凡旋转稳定子。

> Example. Count permutations of eight labels having two cycles of length three and one cycle of length two.

> 例题：求八个标签的置换中，恰有两个三循环及一个二循环的个数。

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Method 5 / 方法 5 [CO01-M05] (continued / 续)

Worked solution

1. Order all eight labels, giving $8!$ descriptions.

2. Each three-cycle has three choices of starting point, the two-cycle has two, and the two three-cycles can be interchanged in $2!$ ways.

3. The count is $8!/(3^2 \cdot 2 \cdot 2!) = 1120$. Cycles are directed permutations, so do not divide for reversal.

例题解析

1. 先排列八个标签，得到 $8!$ 个描述。

2. 每个三循环起点有三种，二循环有两种，两个三循环还可用 $2!$ 种方式交换。

3. 所以计数为 $8!/(3^2 \cdot 2 \cdot 2!) = 1120$。置换循环有方向，不能按反向再除二。

A map on a finite set that is onto is a permutation; an iterate equal to identity restricts every cycle length.

有限集到自身的满射是置换，某次迭代为恒等会限制每个循环的长度。

Practice / 配套练习: AIME-2011-I-Q05 (WB p. 618), AIME-2012-II-Q14 (WB p. 701), AIME-2014-II-Q13 (WB p. 747), AIME-2026-I-Q07 (WB p. 628)

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Method 6 / 方法 6 [CO01-M06]

Count category patterns before permuting within categories

先计类别模式，再排类内标签

Procedure

1. Replace actual labels by residues, low/high classes, parity classes, or repeated-position classes.

2. Derive the allowed category patterns from the original local or global conditions.

3. For each pattern multiply the independent within-class label permutations; check whether all patterns have the same weight.

通用步骤

1. 将实际标签替换为剩余类、高低类、奇偶类或重复位置类。

2. 由原局部整体条件推导允许的类别模式。

3. 对每个模式乘独立类内标签排列数，并检验各模式是否同权。

> Example. Use digits $0, 1, 2, 3, 4, 5$ once in six labeled slots. Slots one, three, five must contain even digits, and slot one cannot be zero. Count the assignments.

> 例题：将数位 $0, 1, 2, 3, 4, 5$ 各用一次填六个有标识位置，第一三五位须为偶数且首位不能为零，求填法数。

Worked solution

1. The category pattern is fixed: even, odd, even, odd, even, odd.

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Method 6 / 方法 6 [CO01-M06] (continued / 续)

2. The first even slot has two nonzero choices; the other two even digits can be permuted in $2!$ ways. The odd digits have $3!$ assignments.

3. The count is $2 \cdot 2! \cdot 3! = 24$. Zero is handled inside the even category.

例题解析

1. 类别模式已固定为偶、奇、偶、奇、偶、奇。

2. 第一偶数位有两个非零选择，其余偶数以 $2!$ 种排列；奇数以 $3!$ 种排列。

3. 共有 $2 \cdot 2! \cdot 3! = 24$ 种，零的限制在偶数类内部处理。

If two histories yield the same repeated-position pattern, count the intrinsic pattern once rather than choosing a repeated pair twice.

若两段构造给出同一重复位置模式，应按内在模式计一次，不能重复选择其见证。

For a regular even polygon, an inscribed rectangle is determined by two diameters. Group opposite vertex pairs by their color type; forbid two solid pairs of the same color rather than testing all rectangles separately. In a residue-constrained array, count residue or category fillings first and multiply the factorials of the actual labels in each class.

正偶数边形的内接矩形由两条直径确定。按对顶点对的颜色类型分类，禁止出现两对同色纯色对，比逐个矩形检查更简。剩余类限制点阵先计类别填法，再乘各类实际标签的阶乘。

Practice / 配套练习: AIME-2006-I-Q06 (WB p. 596), AIME-2011-I-Q05 (WB p. 618), AIME-2011-II-Q14 (WB p. 509), AIME-2012-I-Q15 (WB p. 510), AIME-2016-I-Q08 (WB p. 589), AIME-2017-I-Q11 (WB p. 796), AIME-2021-II-Q03 (WB p. 623), AIME-2022-I-Q09 (WB p. 742), AIME-2023-II-Q07 (WB p. 626), AIME-2025-I-Q05 (WB p. 495), AIME-2025-I-Q10 (WB p. 627), AIME-2026-I-Q09 (WB p. 763)

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Method 7 / 方法 7 [CO01-M07]

Count constrained injections with nested choices or matchings

计算嵌套选项与匹配限制的单射

Procedure

1. Model an assignment as an injection or a perfect matching between two labeled sets.

2. When allowed sets are nested, process the smallest allowed set first and subtract the previously used targets.

3. When one intersection or exceptional match controls feasibility, condition on whether it occurs and prove its representation is unique.

通用步骤

1. 把分配建模为两个有标识集合之间的单射或完美匹配。

2. 允许集合嵌套时，先处理最小集合，依次扣除已用目标。

3. 若一个交点或例外匹配决定可行性，按其是否出现分类，并证明表示唯一。

> Example. Three labeled tasks need distinct machines. Their allowed machine sets are $\{1,2\},\{1,2,3\},\{1,2,3,4,5\}$. Count assignments.

> 例题：三个有标识任务须用不同机器，允许机器集依次为 $\{1,2\},\{1,2,3\},\{1,2,3,4,5\}$，求分配数。

Worked solution

1. Assign the first task in two ways.

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Method 7 / 方法 7 [CO01-M07] (continued / 续)

2. Its machine lies in the second allowed set, so the second task has $3-1=2$ choices. Both used machines lie in the third allowed set, leaving $5-2=3$.

3. The count is $2\cdot 2\cdot 3=12$; nesting makes the continuation counts independent of the chosen machines.

例题解析

1. 第一任务有两种选择。

2. 其机器必在第二允许集中，所以第二任务有 $3-1=2$ 种；已用两机器都在第三集中，故第三有 $5-2=3$ 种。

3. 共有 $2\cdot 2\cdot 3=12$ 种，嵌套性保证后继数与具体选择无关。

A matching counted through a removable object is counted once only if that removable object is unique.

通过可删除物体计算匹配时，只有该物体唯一，表示才不会重复。

Unrestricted perfect matchings on $2n$ distinct items number $(2n-1)!!$: pair a fixed item first, then recurse. For a left-right matching, use $n!$. In a hand covering all rows and columns of a card grid with exactly one repeated row and column, removability is equivalent to presence of their intersection card; establish that it is uniquely removable before counting through the remaining matching.

二 n 个不同物体的无向完美匹配有 $(2n-1)!!$ 个：先为固定物体选搭档，再递推。左右两组之间的匹配则用 $n!$。若牌组覆盖所有行列，且恰一行与一列重复，则可删除性等价于它们交点牌出现；须证明该牌唯一可删，才能通过其余匹配计数。

Practice / 配套练习: AIME-2012-I-Q03 (WB p. 619), AIME-2016-II-Q13 (WB p. 779), AIME-2017-II-Q09 (WB p. 762), AIME-2025-I-Q07 (WB p. 743)

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Method 8 / 方法 8 [CO01-M08]

Insert an extreme element and prove deletion is the inverse

插入极值元素并证明删除为逆过程

Procedure

1. Remove the largest or smallest label and prove the remaining object stays legal.

2. Classify every slot in which reinserting that label preserves the condition.

3. If every smaller object has a fixed number of slots, form a recurrence; if multiple deletions witness the same object, correct their multiplicities.

通用步骤

1. 删除最大或最小标签，并证明剩余对象仍合法。

2. 分类重新插入时所有保持条件的位置。

3. 若每个较小对象具有相同位置数，就建递推；若同一对象有多个可删见证，则修正重数。

> Example. Count permutations of $1, \ldots, 5$ for which, for each k, the entries $1, \ldots, k$ occupy consecutive positions.

> 例题：求 $1, \ldots, 5$ 的排列数，使对每个 k，标签一至 k 都占据连续位置。

Worked solution

1. Begin with the one-entry row containing one. To preserve the condition, each new largest label must be attached to the left or right end.

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Method 8 / 方法 8 [CO01-M08] (continued / 续)

2. Deleting the largest label reverses this operation uniquely, so the construction neither misses nor repeats a permutation.

3. There are two choices at each of four insertions, giving $2^4 = 16$.

例题解析

1. 从只含一的行开始，为保持条件，每个新增最大标签只能附在左端或右端。

2. 删除最大标签唯一逆转此操作，所以构造完备且不重复。

3. 四次插入各有两种选择，共 $2^4 = 16$ 种。

Do not infer a recurrence from legal insertion alone; deletion must preserve legality and recover every object.

仅有合法插入还不足以得到递推，删除也须保持合法并覆盖每个对象。

For permutations with no proper prefix equal to an initial value set, split every unrestricted permutation at its earliest closed prefix. If c_k counts legal first blocks, then $n! = \sum_{k=1}^{n} c_k (n-k)!$. The word earliest is what gives a unique decomposition.

对不允许任何真前缀恰为初始值集合的排列，将任意排列按最早闭合前缀切开。若 c_k 计合法首块，则 $n! = \sum_{k=1}^{n} c_k (n-k)!$；“最早”保证分解唯一。

Practice / 配套练习: AIME-2015-II-Q10 (WB p. 673), AIME-2018-II-Q11 (WB p. 685), AIME-2020-I-Q05 (WB p. 622)